COMPLETE ENGINEERING, COMPONENT MASTERY, & CURRENT ENGINEER NETWORK DIRECTORY

A corrected quick-reference handbook for Elite Dangerous ship engineering on the current Live/Odyssey game. It covers the 20 core-region ship Engineers, including the Witch Head outpost, their current unlock paths, and the required donations or hand-ins.

1. The Engineering Access Protocol

Ship Engineer access normally follows four stages: **Discovery** (public listing or a referral), **Invitation** (career, reputation, permit, distance, or activity requirement), **Unlock** (a donation or hand-in at the workshop), and **Reputation** (raising workshop access to higher grades). A referral is not an unlock by itself.

CURRENT-DATA NOTE

For referral-based Engineers, reach **Grade 3 access plus roughly one-third of the progress bar toward Grade 4** with the prerequisite Engineer to receive the referral. Always use the exact value shown in your in-game invitation where bond totals can vary.

Complete Core-Region Ship Engineer Registry (20)

Engineer & location	Key workshop specialties	How to unlock — current Live/Odyssey
Felicity Farseer Farseer Inc, Deciat	G5 FSD; G5 Detailed Surface Scanner; G3 Thrusters, Sensors	**Referral:** None. **Meet:** Exploration rank *Scout*. **Unlock:** donate **1 Meta-Alloy**.
Elvira Martuuk Long Sight Base, Khun	G5 FSD; G3 Shields; G3 Thrusters	**Referral:** None. **Meet:** travel at least **300 ly** from career start. **Unlock:** donate **3 Soontill Relics**.
The Dweller Black Hide, Wyrd	G5 Power Distributor; G4 Pulse Laser; G3 Beam/Burst Lasers	**Referral:** None. **Meet:** sell commodities at **5 unique Black Markets**. **Unlock:** pay **500,000 CR**.
Tod “The Blaster” McQuinn Trophy Camp, Wolf 397	G5 Multi-Cannon and Rail Gun; G3 Fragment Cannon; G2 Cannon	**Referral:** None. **Meet:** earn more than **15 bounty vouchers**. **Unlock:** hand in **100,000 CR** in bounty vouchers.
Liz Ryder Demolition Unlimited, Eurybia	G5 Missile Rack/Torpedo; G1 Armour Reinforcement	**Referral:** None. **Meet:** *Cordial* or higher with **Eurybia Blue Mafia**. **Unlock:** donate **200 Landmines**.
Juri Ishmaak The Jet’s Hole, Giryak	G5 Sensors and Detailed Surface Scanner	**Referral:** Felicity. **Meet:** claim more than **50 Federal combat bonds**. **Unlock:** hand in the Federal bond value stated in your invitation *(reported as 100,000 or 1,000,000 CR)*.
Lei Cheung Trader’s Rest, Laksak	G5 Shield Generator and Detailed Surface Scanner; G3 Shield Booster	**Referral:** The Dweller. **Meet:** trade at **50 unique commodity markets**. **Unlock:** donate **200 Gold**.
Marco Qwent Qwent Research Base, Sirius	G4 Power Plant; G3 Power Distributor	**Referral:** Elvira. **Meet:** obtain **Sirius permit**. **Unlock:** donate **25 Modular Terminals**.

CURRENT ENGINEER NETWORK DIRECTORY

Complete Core-Region Ship Engineer Registry (continued)

Engineer & location	Key workshop specialties	How to unlock — current Live/Odyssey
Professor Palin Palin Research Centre, Arque	G5 Thrusters; G3 FSD	**Referral:** Marco Qwent. **Meet:** reach a maximum distance of **5,000 ly** from career start. **Unlock:** donate **25 Sensor Fragments**.
Chloe Sedesi Cinder Dock, Shenve (Witch Head)	G5 Thrusters; G3 FSD	**Referral:** Marco Qwent. **Meet:** maximum distance of **5,000 ly** from career start. **Unlock:** donate **25 Sensor Fragments**.
Zacariah Nemo Planetary Lab, Yoru	G5 Fragment Cannon; G2 Multi-Cannon	**Referral:** Elvira. **Meet:** *Friendly* with **Party of Yoru** and obtain its invitation. **Unlock:** donate **25 Xihe Biomorphic Companions**.
Lori Jameson Jameson Base, Shinrarta Dezhra	G5 Sensors, Life Support, Fuel Scoop	**Referral:** Marco Qwent. **Meet:** Combat rank *Dangerous* and the **Founders World permit**. **Unlock:** donate **25 Kongga Ale**.
Selene Jean Prospector's Rest, Kuk	G5 Hull Reinforcement and Armour	**Referral:** Tod McQuinn. **Meet:** mine and refine **500 tonnes of ore**. **Unlock:** donate **10 Painite**.
Bill Turner Turner Metallics Inc, Alioth	G5 Plasma Accelerator; G3 Sensors/Life Support	**Referral:** Selene. **Meet:** *Friendly* with the Alliance and an **Alioth permit**. **Unlock:** donate **50 Bromellite**.
Didi Vatermann Vatermann LLC, Leesti	G5 Shield Booster; G3 Shield Generator	**Referral:** Selene. **Meet:** Trade rank *Merchant*. **Unlock:** donate **50 Lavian Brandy**.
Hera Tani The Jet's Hole, Kuwemaki	G5 Power Plant and Detailed Surface Scanner	**Referral:** Liz Ryder. **Meet:** Imperial Navy rank *Outsider*. **Unlock:** donate **50 Kamitra Cigars**.

CURRENT ENGINEER NETWORK DIRECTORY

Complete Core-Region Ship Engineer Registry (continued)

Engineer & location	Key workshop specialties	How to unlock — current Live/Odyssey
Broo Tarquin Broo's Legacy, Muang	G5 Beam, Pulse, and Burst Lasers	**Referral:** Hera Tani. **Meet:** Combat rank *Competent*. **Unlock:** donate **50 Fujin Tea**.
Tiana Fortune Fortune's Loss, Achenar	G5 Limpet Controllers and Scanners	**Referral:** Hera Tani. **Meet:** *Friendly* with the Empire and the **Achenar permit** *(Imperial Navy rank Squire or higher)*. **Unlock:** provide **50 Decoded Emission Data**.
Ram Tah Phoenix Base, Meene	G5 Chaff, Heat Sink, and Point Defence	**Referral:** Lei Cheung. **Meet:** Exploration rank *Surveyor*. **Unlock:** provide **50 Classified Scan Databanks**.
Colonel Bris Dekker Dekker's Yard, Iapetus, Sol	G5 Beam Laser; G4 Sensors; G3 Shield Generator	**Referral:** Juri. **Meet:** *Friendly* with Federation plus **Sol permit** *(Federal Navy Petty Officer)*. **Unlock:** hand in the Federal bond value in your invitation *(reported as 1M or 10M CR)*.
The Sarge The Beach, Beta-3 Tucani 2 b a	G5 Hull Reinforcement and Armour	**Referral:** Juri. **Meet:** Federal Navy rank *Midshipman*. **Unlock:** provide **50 Aberrant Shield Pattern Analysis**.

Engineer branch map

Felicity → Juri → Colonel Bris / The Sarge
Elvira → Marco → Professor Palin / Chloe Sedesi / Lori Jameson
Elvira → Zacariah Nemo
The Dweller → Lei Cheung → Ram Tah
Tod McQuinn → Selene Jean → Bill Turner / Didi Vatermann
Liz Ryder → Hera Tani → Broo Tarquin / Tiana Fortune

Why this edition differs from the old guide

This corrected edition removes Chesterfield Flandry from the normal ship-engineer roster and restores the missing core-region Engineers, **Colonel Bris Dekker** and **The Sarge**. It also corrects high-impact errors such as Elvira's 3 Soontill Relics, Selene's 10 Painite, Tiana's Hera referral, and Zacariah's Xihe Biomorphic Companions donation.

■■ COMPONENTS, SCO, & ODYSSEY ENGINEERING NOTES

2. Component Classes & Ratings

A module's **class** is its physical size; its **rating** describes the module's performance profile. Ratings do not "conflict"—they are trade-offs. Select them based on your ship's role, mass budget, heat, integrity, and power draw.

Rating	Practical meaning
A-rated	Usually the highest-performing option; common for FSDs, power plants, distributors, shields, and combat-critical modules.
B-rated	Very high integrity but heavy. A specialist survivability choice, not a general-purpose upgrade.
C-rated	Middle-of-the-road performance and cost.
D-rated	Lowest mass. Often ideal for non-critical exploration modules because reduced mass improves jump range.
E-rated	Entry-level stock equipment; replace selectively as your budget and ship role require.

3. Supercruise Overcharge (SCO) FSDs

SCO-capable Frame Shift Drives are purchasable modules that permit Supercruise Overcharge. They can greatly reduce long in-system travel times, but sustained overcharge consumes fuel quickly and creates severe heat. Compare the exact class/rating in your outfitting screen or build planner; do not assume every SCO drive has the same jump performance.

Practical SCO routine: fit a heat sink if your ship runs hot, use short bursts, watch fuel and temperature, and stop before heat or fuel becomes a safety problem.

4. Guardian & Remote Engineering

The Guardian FSD Booster is unlocked once through a Guardian Technology Broker and then can be purchased repeatedly for credits. The largest variant adds a flat **+10.5 ly** jump-range bonus. You may pin one blueprint per Engineer for remote application at stations, but Experimental Effects still require a visit to an appropriate Engineer workshop.

5. Odyssey On-Foot Engineers

Odyssey suit and weapon engineering is separate from ship engineering and is not included in the 20-Engineer ship registry. The core-region entry branches begin with **Hero Ferrari**, **Jude Navarro**, and **Domino Green**; their referrals lead to Wellington Beck/Uma Laszlo, Terra Velasquez/Oden Geiger, and Kit Fowler/Yarden Bond respectively. Their unlocks use Odyssey materials and mission or activity requirements, not ship-engineer reputation grades.

■ CRITICAL OPERATIONS NOTE

Before engineering, validate the complete build in Coriolis, EDSY, or another current planner: changes to power draw, thermal load, mass, shield strength, and weapon distributor demand can make an apparently valid modification impractical in flight.

Source basis: current Elite Dangerous Field Manual engineer pages and unlock guides, cross-checked against INARA's engineer directory. Values that current references identify as variable are explicitly marked "use your invitation." This is a gameplay quick reference, not an official Frontier publication.